Freddy’s personal devlog

Subject	Comp 208
Date	@September 30, 2025
Keywords	

Name
Worksheet 1 Braitenburg
Worksheet 2 Finite State machine
Worksheet 3 Odometry
Worksheet 4: Using Motors
Worksheet 5: Control systems
Worksheet 6: Making a mechanism
Worksheet 7: Sensors
Worksheet 8: Following and sensing
Worksheet 9: Pathfinding

To do list:

CURRENTLY DOING:

Practice encoders and getting brushed motors working.

Get wheels

Get box

Make PCB

Start looking into camera technology

Battery

Research

For prototype

Wheels

2 wheel drive

2 motors

Light weight

PCB for the actual working product
Box
Wheels

DAY 1 – 22/09/2025

Activity: Attended Matt's lecture and prepared for upcoming work.

Reflection: A lot to take in a think about for the future. Need to not too far ahead and focus on little goals.

Next Steps: Research autonomous systems and cybernetics.

DAY 2 – 23/09/2025

Activity: Module workshop with Ben; learned lab rules (lone working, clear desk, lockers).

Reflection: Inspired a nightlight robot idea: moves to dark spots and activates automatically.

Next Steps: Explore feasibility of mobile nightlight robot.

DAY 3 – 27/09/2025

Activity: Group assignment confirmed: working with Ryn and Jareth. Brainstormed ideas for "scrabbling" contract (rugged outdoor robot that can locate a can).

Reflection: Key tasks: transport outdoors, recognize target tasks, calibrate distance, know when task is complete. Current idea: Roomba-style robot.

Next Steps: Prepare sketches and logic flow for outdoor operations.

DAY 4 – 28/09/2025

Activity: Discussed movement base with Jareth (treads, wheels, flight). Attended Matt's lecture on human-robot connection.

Reflection: Interface affects attachment to robots; naming them could increase meaning. Our contract has minimal interface, but it could be included as a final measure.

Next Steps: Decide movement system; considered UI for robot. Multiple LEDS that communicate different problems.

DAY 5 – 30/09/2025

Activity: Planning session with Jareth; finalized primary concepts:

- Lightweight, nimble, torque-heavy robot
- Spiked treads
- Finds a can and returns to start
- Battery powered (10V)
- Robust single-purpose code, alerts system failures with beepers
- Low-resolution camera for color detection
- Secondary snake with wheels idea to explore later. So less power but more wheels and more traction.
- Reflection: Prioritized robustness and outdoor traction.
- Next Steps: Prototype movement system and integrate sensors.

DAY 9 – 09/10/2025

Activity: Presented ideas in 208 tutorial; pitched project to Matt. Workshop paired with Jareth. Improved teamwork and task delegation.

Reflection: Braitenburg machine initially imprecise; using DC programmable power supply improved functionality. Developed initial Finite state machine with LDR detecting light to switch directions. IR may improve precision.

Next Steps: Replace LDR with IR sensors for more accurate object detection.

DAY 19 – 23/10/2025

Project Update:

Presented the prototype to Matt; received valuable feedback highlighting key areas to focus on. With the power issue resolved, the next major pushes are:

- Researching and improving the steering system.
- Exploring a Jareths four-wheel drive setup with a back step focus on steering columns.

I've been researching motorcycle chariot front-end designs using dual push-pull rods. Goal: keep the system simple but evaluate alternatives to make an informed decision.

DAY 21 – 29/10/2025

Mechanical & Electrical Assembly:

- Components: 4 brush motors, 1 Arduino Nano, 1 H-Bridge, 1 camera, 1 box, 1 12V battery.

Technical Parameters:

Given/Primary:

- Mass m=18.65 kgm
- Wheel radius r=0.0508
- Linear speed v=0.10 m/sv
- Gearbox ratio G=63:1G
- Driven wheels n=2n or 2n=2 differential drive
- Torque safety factor S=3S

Assumed/Explicit:

- Gravity g=9.81m/s²
- Worm-gear efficiency n=0.6\eta
- Terrain coefficients: concrete 0.01, gravel 0.02, grass 0.06

Torque and power calculations for terrain & slope:

Case	Frr (N)	Fhill (N)	Ftotal (N)	twheel (N·m)	Pwheel (W)	tmotor (N·m)	Pmotor (W)
Concrete	1.83	0	1.83	0.139	0.274	0.0037	0.457
Gravel	3.66	0	3.66	0.279	0.549	0.0074	0.915
Grass	10.98	0	10.98	0.836	1.652	0.0221	2.744

Housing: Rectangular box (8 inches) using Arduino packaging box; sufficient for all components and pre-drilled holes.

Localization & Data Management:

- Camera module for visual localization, capturing images of start and target locations.
- Multiple color-coded LEDs indicate operational state or events (e.g., red LED for camera activity).

FSM Design:

State	Trigger/Entry	Exit/Transition
Initialization	Power switch	Calibration complete
Idle or Standby	Awaits command	Navigation starts
Navigation	Object search	Object found start Visual Tracking
Visual Tracking	Camera detects target	Object located start Navigation return

Obstacle Avoidance	Sensor triggered	Clear path begin set directions
Error o r Fault	Fault detected	Resolved, go back to Idle
Shutdown or Stop	Task completed	Power off

Motor Control: H-Bridge with Arduino Nano; integrated encoder readings.

Data Structures Example:

```
enum RobotState {
    INIT,
    IDLE,
    NAVIGATION,
    VISUAL_TRACK,
    AVOIDANCE,
    ERROR,
    SHUTDOWN
};

enum ErrorType {
    OK,
    SENSOR,
    MOTOR,
    OVERCURRENT,
    COMMS
};

struct RobotParams {
    float wheelRadius;
    float wheelBase;
    float gearRatio;
    float encoderTicks;
    float ticksPerRevolution;
    uint8_t drivenWheel;
};

struct RobotStatus {
    RobotState currentState;
    ErrorType lastError;
    uint8_t ledStateMask;
    long pos;
    long lastPos;
    unsigned long lastRPMTime;
    float rpm;
    bool isRaising;
};
```

Software Architecture:

- Modules: main.ino, motor.cpp/h, camera.cpp/h, trajectory.cpp/h, robot.h
- FSM orchestrates robot’s behaviors; camera actively searches for object, locomotion optimizes view.
- Upon completion, robot reverses path to starting location.

Error Handling & Edge Cases:

- LEDs indicate error states.

- Edge cases (robot stuck, lost object, unusual obstacle) trigger predefined movement patterns, e.g., Fibonacci or vector search, to attempt recovery.

DAY 22 - 03/11/2025

Whilst Jareth worked on refining and charging the finite state machine, I worked on everything outside of the machine. Such as the Huffman table. I had a bit of trouble wrapping my head around it so it might be incorrect.

States	NONE	RIGHT IR	LEFT IR	BOTH	OUTPUTS
	0,0	0,1	1,0	1,1	
1: Stop	2/F	4/L	3/R	1/S	F:MOTORS FORWARD
2: Forward	2/F	1/S	1/S	2/F	R:MOTORS ROTATE RIGHT
3: Rotate Right	1/S	1/S	1/S	4/L	L:MOTOR ROTATE LEFT
4: Rotate Left	1/S	4/L	3/R	3/R	S:MOTORS STOP

- Went to Matts lecture about actuators. Recounting the 4 key actuation systems: Pneumatic, Hydraulic, Fuel actuators, Electric motors.
- Pneumatic using pressurised water or air driven in tubing.
- Hydraulic using heavy air compression around tubing.
- Fuel actuators uses a system that controls amount of and pressure of fuel.
- Electric motors - Typically controlled by a shaft between magnetic forces and shifting between those forces.

Turned out that was last weeks lecture.

This weeks lecture was about control theory. The whole system the idea we take something out of the robot put something back in to see what we need to change in the future.

Control Periods, 0.5cm is a good control period baseline to build upon.

Control Algorithms things such as BANG BANG CONTOROL: Between the desired value and measured value is the error.

PID CONTROL SYSTEMS: consider the present, remember the past, predict the future.

Proportional Control: reduce your speed more as you get closer to the reference distance.

- This weeks lecture involved feedback processes. How we can can act on the feedback we gather.

Day 23 04/11/2025

Tutorial with Ben: I worked on starting the PCB, Lab Bot, or more specifically the motors on the labbot and being able to practice programming a moving object. It's more about getting acquainted with the technology to ease myself in 2 weeks before the prototype.

Day 24 - 09/11/2025

Easy eda for the pathfinding robot.

Goal make a PCB for the robot.

Outcome: Only have attachments for motors and a battery.

Going forward: putting in more attachements for camera use and 2 more motors. or 1 with a drive chain.

I restarted my sketch this time with better routing and tried it with a smaller board to find that a 200mm^2 board is probably best since it will leave plenty of space for future attachments and batteries. This is what I have so far.

https://github.falmouth.ac.uk/JH321562/Comp-208-Brab-Putt/tree/main/CAD

DAY 25 - 10/11/2025

Goal: attended Matts lecture on mechanisms and materials, start on poster for the assignment, work on the labbot for worksheet 5.

Outcome: Theorizing Implementation of long lasting principles into my own devices. Such as gearing. bearings and brushes, cams and followers, springs, assemblies.

Started work on poster, added it to the repository, made an abstract to get started that will be rewritten.

As far as the labbot I worked on odometry code, adding it to my PID control code and the labbot controls code. It's too long for a code snippet so I've attached the url to the github:

https://github.falmouth.ac.uk/JH321562/Comp-208-Brab-Putt/tree/main/freddy_practice/encoderpract_copy_20251110134017

Going forward I will have to learn about using h files since the file is over a hundred lines and it will keep building so to make it more readable I will start moving functions over to ensure its readability and ease of understanding. Will try to implement long lasting principles into my own creations.

Learnt about Pierre Jacquet Droz 1773 robot that is now 250 years old. An autonomation that could last over 250 years a quarter of a century. One of the first instances of the first programmable device that is still functioning as it would when it was first created.

DAY 26 11/11/2025

Goal: Finish PCB, clear nets, ensure it's able to be built upon and doesn't later hinder us.

Outcome: I added a mosfet, a motor to the PCB and cleared most of the nets.

Going forward: It is too soon to be get the PCB since it's early in the project to have something so definite. Especially since Jareth or Ryn isn't in and it won't be agreed upon. I will continuously work on it.

Working on the PCB for the second time. Reading about encoders to try and get my encoder code to function for the labbot and hopefully our device, since they're using very similar technology.

https://adityakamath.github.io/2021-12-12-getting-started-with-odometry

From this I found a useful equation for programming these kind of robots. From that equation I put it into arduino code that I could hopefully use in my current arduino for movement: https://github.falmouth.ac.uk/JH321562/Comp-208-Brab-Putt/blob/main/freddy_practice/encoderpract_copy_20251110134017/encoderpract_copy_20251110134017.ino

```
float angular_velocityw1 = 0;
float angular_velocityw2 = 0;
float angular_velocityw3 = 0;
float angular_velocityw4 = 0;
float radius = 0;
float system_dimensionx = 0;
float system_dimensiony = 0;
float longitudinal_velocity = (angular_velocityw1 + angular_velocityw2 + angular_velocityw3 + angular_velocityw4) * (rad
ius / 4);
float transveral_velocity = (-angular_velocityw1 + -angular_velocityw2 + angular_velocityw3 - angular_velocityw4) * (rad
ius / 4);
float angular_velocity = (-angular_velocityw1 + angular_velocityw2 - angular_velocityw3 + angular_velocityw4) * (radiu
s/4 * (system_dimensionx + system_dimensiony));
void setup()
{
  // put your setup code here, to run once:

}

void loop()
{
  // put your main code here, to run repeatedly:

}
```

To find the radius, I would have to measure each wheel, to find the angular velocities I would have to measure the rate at which each wheel is spinning using the encoder (revolutions over time). For the system_dimension x & y I can find them by

physical measurements of the scrabbling robot or LABBOT in this case.

DAY 27 12/11/2025

Goal: Incorporate the equation above into the encoder code for the odometry task.

Outcome: Unfortunately I'm currently unable to test the code since my debugger won't allow the encoder library to upload this to an raspberry pi.

Going forward: Researching more ways that precision can be used in moving robots. Gaining deeper insights into motors and wheels.

Keeping the original equations and changing the declarations to what actually fitted the previously established code I managed to implement the equation to help with accuracy and tuning in both worksheet 5 and the scrabbling device.

```
void velocity()
{
  static long lprevticks = 0;
  static long rtrevticks = 0;
  static unsigned long lasttime = 0;

  unsigned long now = millis();
  float deltatime = (now - lasttime) / 1000.0;
  if (deltatime <= 0)
  {
    deltatime = 0.0001;
  }

  long lticks = myEncoderL.read();
  long rticks = myEncoderR.read();

  long ldeltaticks = lticks - lprevticks;
  long rdeltaticks = rticks - rprevticks;

  float wheelradius = 0.00;
  float angular_velocityw1 = (ldeltaticks / ticksPerDegree) * (PI / 180.0) / dt;
  float angular_velocityw2 = (rdeltaticks / ticksPerDegree) * (PI / 180.0) / dt;

  float longitudinal_velocity = (wheelradius / 2.0) * (angular_velocityw1 + angular_velocityw2);
  float angular_velocity = (wheelradius / axisWidth) * (angular_velocityw2 - angular_velocityw1);
  lprevticks = lticks;
  rprevticks = rticks;
  lasttime = now;
}
```

Day 28 14/11/2025

Researching armature resistance control method

Goal: Gain deeper understanding of how else the motors could be controlled, for higher capabilites and further fault proofing.

Outcome: I learnt that increasing the resistance causes a larger voltage drop, which lowers the armature voltage, armature current, and motor speed. Which helps to control the speed.

Going forward: Using this method, to help control the vehicle would allow for non-user controlled capacity. It can be controlled by taking readings from itself.

here's what I found: ""

The concept of speed regulation is different from the speed control. In speed regulation, the speed of the motor changes naturally whereas in dc motor the speed of the motor changes manually by the operator or by some automatic control device. The **speed** of DC Motor is given by the relation shown below:

Here equation (1) shows the speed is dependent upon the supply voltage V, the armature circuit resistance Ra and the field flux ϕ, which is produced by the field current.

https://circuitglobe.com/speed-control-of-dc-motor-armature-resistance-control-and-field-flux-control.html

DAY 29 15/11/2025

Goal: wire up the internals of the prototype.

2 motors 1 h bridge 1 arduino.

Outcome: followed MATTS diagram from the odometry worksheet and adapted it for 2 motors, then set the variables on my encoder code to fit the same ports on the arduino.

Going forward: I will begin testing on the internals ensuring they work reliably and that my wiring and code functions as expected.

I wish I could of done more than just wire something up, especially with another presentation day coming up and how much I want to test this code, but with dealing with higher voltages in a weary state meant I should make the grown up decision and call it there for the day.

DAY 30 16/11/2025

Finished wiring up and checking it's all in the right ports.

Goal: Final evaluation and showcase to the rest of the group.

Outcome: I have a lot more confidence in my code and mechanism.

Going forward: I will test it tomorrow during my free 4 hours in the lab.

I did want to test it today however I was unable to or at least it wouldn't of been safe since the connections have been soldered however without rubber on top so it's exposed with 12 volts going through. I want to reduce the risk of a short so I left it to Monday to test it practically and not theoretically.

DAY 31 17/11/2025

Goal: Get two motors moving, carrying on from yesterday.

Outcome: Shorted my laptop because one of the connector pins were set in a different order to the other motor, which I hadn't realised.

Going forward: I will test it on a lab laptop then if it works continue on my laptop. One motor was working so now that I have correctly wired it, it should function.

I covered up the soldered ends, which I ended up reshouldering so that the ends matched the other motor, and I had a level of continuity. One motor was working and I couldn't figure out why the other wasn't so I i tried the h-bridge and switching the out ports which led to a short on my laptop. WHen trying to find what caused this Maya had pointed out that the other motor was set up differently but with the same colored wires. Only 3 were in a different order.

I also made some test code:

attachment:3465202d-adcd-4199-9ff2-349fb9987152:MicrosoftTeams-video_(1).mp4

```
// Motor pin declaration
int motorA = 8;   // Right motor IN1
int motorB = 9;   // Right motor IN2
int motorC = 10;  // Left motor IN3
int motorD = 11;  // Left motor IN4
```

```
void one()
{
  analogWrite(motorA, 200);
  analogWrite(motorB, 0);
  analogWrite(motorC, 200);
  analogWrite(motorD, 0);
}
void two()
{
  analogWrite(motorA, 0);
  analogWrite(motorB, 200);
  analogWrite(motorC, 200);
  analogWrite(motorD, 0);
}
void three()
{
  analogWrite(motorA, 0);
  analogWrite(motorB, 200);
  analogWrite(motorC, 0);
  analogWrite(motorD, 200);
}
void four()
{
  analogWrite(motorA, 200);
  analogWrite(motorB, 0);
  analogWrite(motorC, 0);
  analogWrite(motorD, 200);
}

void setup()
{
  // put your setup code here, to run once:
  Serial.begin(9600);
  pinMode(motorA, OUTPUT);
  pinMode(motorB, OUTPUT);
  pinMode(motorC, OUTPUT);
  pinMode(motorD, OUTPUT);
}

void loop()
{
  // put your main code here, to run repeatedly:
  one();
  delay(2000);

  two();
  delay(1000);

  three();
  delay(2000);

  four();
```

```
  delay(1000);
}
```

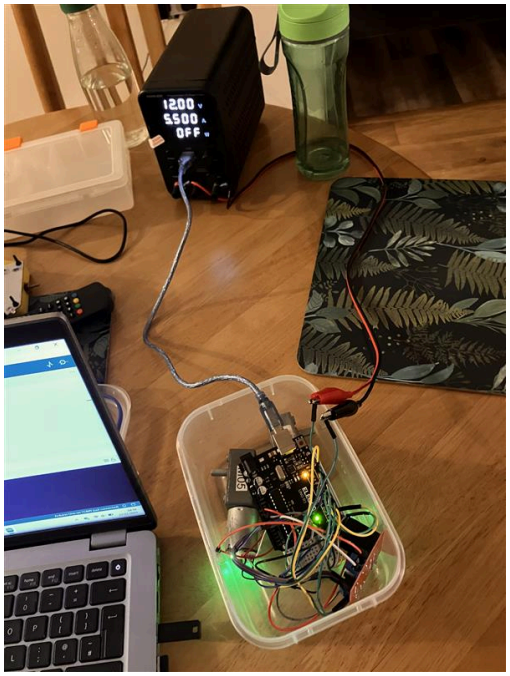
Went to Matts Lecture where I learnt about:

Sense and sensors, completing the cybernetic loop. Passive and active sensors, define ranges, resolution, precision, and accuracy of sensors. Active sensors do .

Machine vision, lidar, ultra sonic, infrared, etc. Their advantages, drawbacks, what applications their used for.

I managed to get both motors working. Turns out the safe way to do it is with the power supply with the code already uploaded.

attachment:6523f002-9061-40c9-8e1d-35f95396d241:MicrosoftTeams-video_(2).mp4



DAY 32 18/11/2025

I managed to get a to get two wooden wheels and a rod inbetween so now it has 2 wheels. I wanted to print off two more wheels as it was a more tapered fit as these would be attached to the motors however orca slicer wouldn't open, so I decided to leave that job to Jareth. I need to work on my code and getting it more simultaneous. However it is close to moving. I also need batteries for the motors.

Because of how I've placed the motors one of them works inversely to the other. So it appears to be going reverse when the other is going forward.

Goal: Get 4 wheels and a moving prototype without battery.

Outcome: I only got 2 wheels, and tested the code and it all fits together nicecly.

Going forward: Get two more wheels, work on code and purchase battery pack.

DAY 33 19/11/2025

attachment:d2ea9747-099e-4de5-ba83-cd405c10f3ad:Video.785259619.354462.mp4

attachment:989d5811-19fa-4d27-9725-979d35650545:original-D7B85FBB-8FC3-465C-9268-049422982DED.mp4

Met with Jareth in the lab, we pulled through and put our work together and made one working prototype. We put together our separate prototypes and with my internals, and Jareth's box and wheel setup, worked instantly.

Goal: Fashion together, a prototype.

Outcome: We have a pulled in prototype, we just need one with a battery so that it can be more portable.

Going forward: We will continue to communicate effectively. Start finalizing our project into something more applicable.

DAY 34 20/11/2025

Goal: New Shell, new wheels, new power for anker battery.

Outcome: Matt was pleased the robot moved well by itself but with 22 days left, we now need to take it further and include machine vision and improve every aspect of the device.

Going forward: Purchase new components, work on set out its faults.

Today we had a demonstration with Matt. Unfortunately I could only get a battery for the Arduino and not for the 12 volt so now looking for answer for transferring power from the one battery to the 4 motors.

States. Finding / searching state / Going back states.

I think for a box, laser cut a box so that it has extremely accurate hole positions for applying the motor.

Work on gearing mechanisms for power application. Power non-directly to wheels.

SO to do list in terms of easiest to hardest:

BOX - 1 day

GEARING - 2 days

new design - 3 days

Battery - 5 days

NO PCB

Machine vision - 2 weeks

If I were to focus on the hardware and Jareth Software, rotating every week then I think we would have a good flow state.

2nd half movement is complete. Now to finish it have all wheel drive. While Jareth looks into machine vision.

Then hopefully by next week we have a moving prototype, that is self controlled.

A prototype that goes to a certain location then moves back.

Then A prototype that can detect certain objects and move towards them and moves back.

Wish list:

Wheels

https://www.amazon.co.uk/dp/B099DTX315/ref=sspa_dk_detail_1?pd_rd_i=B099DTX315&pd_rd_w=Kyiz9&content-id=amzn1.sym.9a64fe05-cdee-4d53-a27b-f3614d726545&pf_rd_p=9a64fe05-cdee-4d53-a27b-f3614d726545&pf_rd_r=J11E9EJCR07D6YXAGZ5H&pd_rd_wg=dcHNR&pd_rd_r=4b0399ae-ca43-4ea6-acee-08dd302433c7&aref=hwd5l0Sxl0&sp_csd=d2lkZ2V0TmFtZT1zcF9kZXRhaWxfdGhlbWF0aWM&th=1&tag=operagx-def-sp-co2-gb-21

Battery converter

https://www.amazon.co.uk/AITRIP-Converter-Adapter-Reduced-Regulator-Type/dp/B0DCGGGPFZ/ref=sr_1_10?crid=2ZD7I85Q1SYNI&dib=eyJ2IjoiMSJ9.41wdIjTOEhxz4IH40iyryZxct72iVWn6XLthmGHprmxxRegTBGizrIvcorIRmQm-CxKWF1VqZF0QImIIlNPBmUXFRSmrfHRZArDu7Qf7juuPBpWXQgirH-RFb0w_wCChyfFS1d2ewEy9LskMYzCk5aAZOtYZE66Tmk3n9yZR1vMfg1xCI6vAw_P1GtanBCKZ4bEJRswtyJOdjRmlOEPP7mn

NH10eJb_k8A2QHZpbAY2Jc3AUmPth4D2c7mvLvJIh0VNf3iimmKeig.xRMj15lUjIaIYnNWKKNbuveJNXVnjO_IqnpN1fJ7R8k&di10

DAY 35 21/11/2025

Goal: Upload videos to devlog / buy components.

Outcome: Today was admin since a lot of my pictures and videos were missing. I had components I needed to purchase to make it to the next stage.

Going forward: parts should arrive tomorrow. I've get straight to work on testing them and using them.

I spent around £80 on wheels,12 volt battery, 1 more encoder motor, and cambelt and gearing for my own personal use, as well as 1 300 rpm dc motor to test my idea. It should me grasp the idea of gear ratios. Me and Jareth agreed to put it on a place plate. The chasis is built to work on it's own without the cover on the top.

DAY 36 22/11/2025

Goal: Test out apparatus.

Outcome: Only the cambelt gearing arrived, so unfortunately I couldn't test out anything. Although I did learn that the gearing doesn't fit the motor.

Going forward: I will test out components and starting by attaching it to encoders.

Only one thing arrived so I was unable to complete what I wanted to. I want to focus on practical building things rather than fixating on code, CAD, CAM. Rapid prototyping, so not getting the parts was a sting.

I've tried to do more labbot code with the new encoder however I still have some kinks to straighten out. It currently isn't compiling yet.

```
void control_left_motor(int motor_power)
{
motor_power = constrain(motor_power, -255, 255);

if (motor_power > 0) {
analogWrite(L_MOTOR_A, motor_power);
digitalWrite(L_MOTOR_B, LOW);
} else if (motor_power < 0) {
analogWrite(L_MOTOR_B, -motor_power);
digitalWrite(L_MOTOR_A, LOW);
} else {
digitalWrite(L_MOTOR_A, LOW);
digitalWrite(L_MOTOR_B, LOW);
}
}

void control_right_motor(int motor_power)
{
motor_power = constrain(motor_power, -255, 255);

if (motor_power > 0) {
analogWrite(R_MOTOR_A, motor_power);
digitalWrite(R_MOTOR_B, LOW);
} else if (motor_power < 0) {
analogWrite(R_MOTOR_B, -motor_power);
digitalWrite(R_MOTOR_A, LOW);
} else {
digitalWrite(R_MOTOR_A, LOW);
digitalWrite(R_MOTOR_B, LOW);
}
}
```

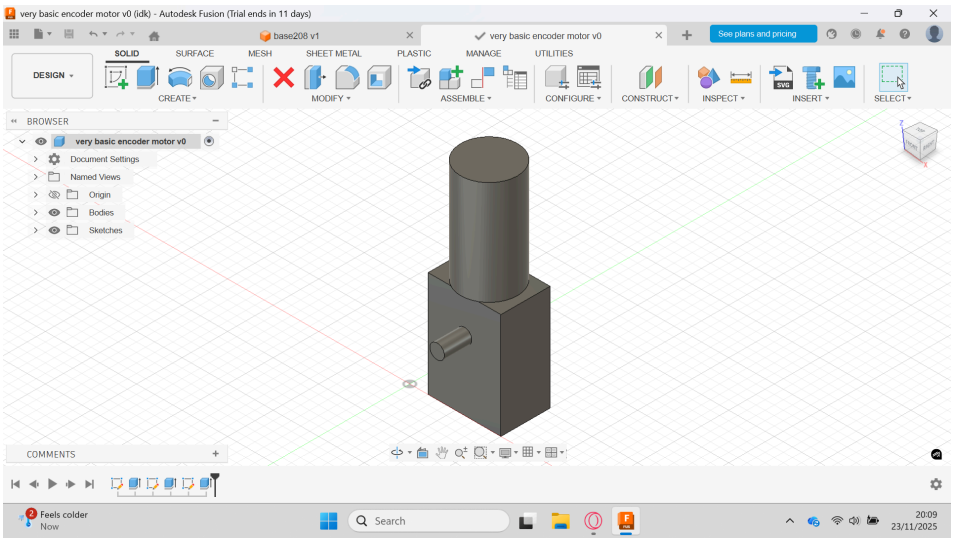
DAY 37 23/11/2025

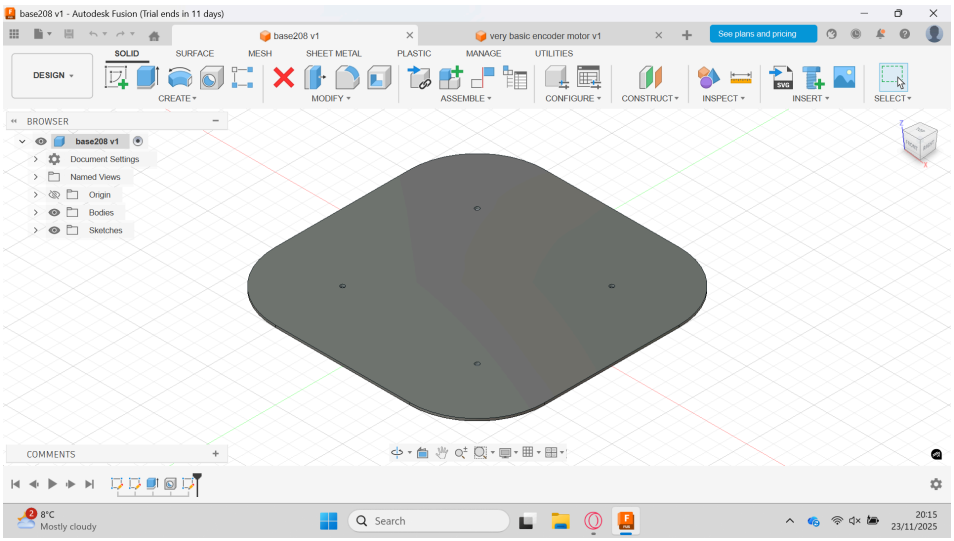
Goal: start testing motors and get things working.

Outcome: Ended up on fusion making encoders and boards to later cut for accuracy.

Going forward: The parts will arrive when they arrive I have to make sure I keep moving forward with the project even if they're not here.

Unfortunately only the battery and other encoder motor turned up today. I'll need to wire up the new encoder motor tomorrow and test it out. This lack of progress and need to complete thing





DAY 38 24/11/2025

Goal: Read up on PS3 machine vision camera.

Outcome: Revan a third year sent me a link to the github repository of a machine vision camera, I had bought:Releases · jkevin/PS3EyeDirectShow. Reading through this and how untested it is led me to think of other ways to approach this. OPENSOURCE.

Going forward: I will look at apps for a phone that can detect an object and send signals to an arduino or a bluetooth controller. I feel that may be the easiest way forward but it's time consuming in research and trial that might not actually work.

Lecture with Matt:

Navigation and localisation.

Identify basic methods: robot navigation (line and wall following), direction and goal, probabilistic methods to find objects in environment. Pledge algorithm.

PID can help control line following to make less erratic.

Dosh dat method, line following on for localising and the other for something…Navigation.

Mediapipe. Check it out.

https://github.com/google-ai-edge/mediapipe

The other parts came - the wheels and other motor to test my idea of two encodes at the front and I DC motor at the back powering the back two wheels.

DAY 39 25/11/2025

Goal: Get board made up / laser cut.

Outcome: Got the components together and the cambelts are the way forward.

Going Forward: We will put the components on the board then create a system on top of the board.

We managed to connect the battery to new ends so that it can be charged in the lab.

Wheels are 4.5 mm hold. 12 mm inside hexagon. I want to make a bolt that goes from the outside, bolt screws into the hexagon the bolt then goes through Jareths box join for finally a bearing on a positioner the one that looks like a little bridge to hold it up then thats the bolt. Inbetween the bearing and the wall is the cambelt that drives it.

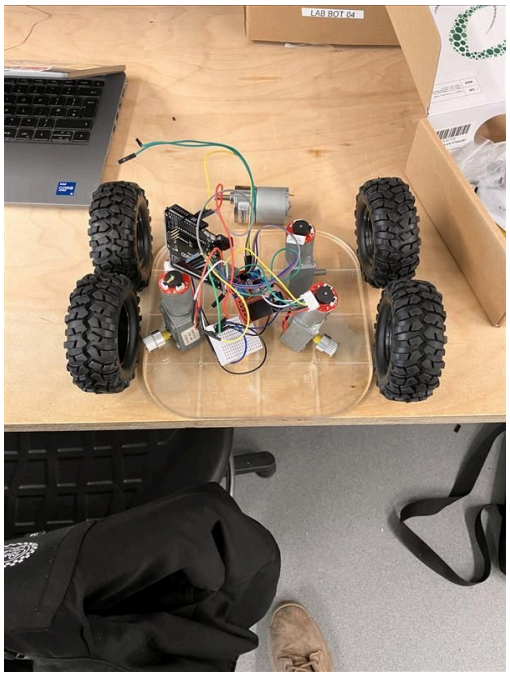
Okay so change of plan. It's just the bearing that look like bridges holding it in place on the board drills into the rod for precisely keeping in the cambelt gearing. Then look at how tall it needs to be for the gearing. Mounted bearings.

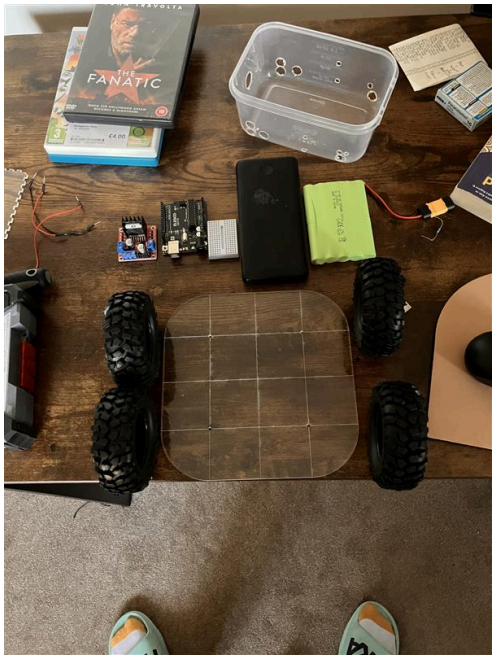
When I got home I began working on placing everything on the board and trying to work out how it would all fit together. Doing this I wanted to just solve it by layering it on top of each other. Laser cutting more boards, which could work and would keep it tidy but it definitely all fit well together I just need to think…Scrutinize about it. At the moment the hardest part is the encoders and how they're always facing a different direction. Not to mention their is 4 encoders although I want 2 and 1 dc motor. I came up with facing the batteries, arduino, h bridge. Well all of the components upwards which gave me a lot of space but means I can't zip tire it down and will need a holder. To 3d print holders. 3d printing and CAD take me more time than I would like and so I would prefer to tire it down but for a flush finished project I might have to.

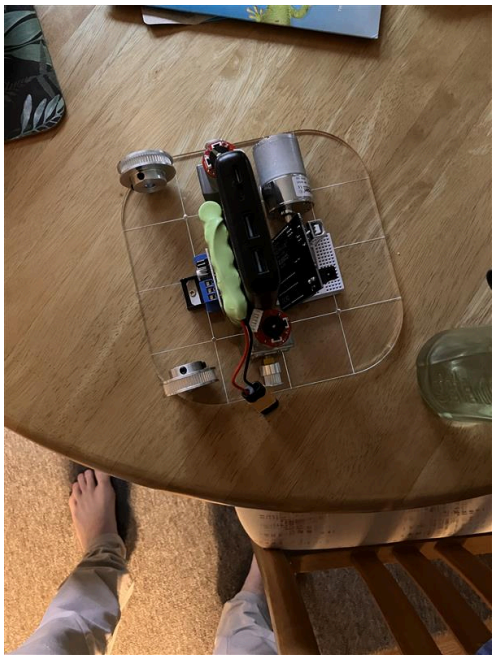


attachment:3787acd5-154a-409d-b756-3136d63b90af:video-20251125-104502-d7d3e3e1.mov

attachment:142a4932-b648-40fc-b827-ad430c6c4385:video-20251125-102431-70a4636f.mov







DAY 40 26/11/2025

Goal: Find the right positioning for everything.

Outcome: I only have decided to the position of the secondary gears that turn the rod.

Going forward: I'll have to speak with Jareth tomorrow and get his input before we start building onto it.

What I'm struggling most with is the encoder motors they have to be fit a very exact way that allows them to be plugged in and in a good location for turning the gears. The problem is the best position for them is the least suitable for everything else. I'm thinking the arduino battery will have to go underneath. It is waterproof afterall. Not to mention space for the camera operation.

Day 41 27/11/2025

make endcap that is at least 27 millimetres.

Goal: Final Redesign

Outcome: I've made an encoder case that is ready to be 3d printed and mounted to the board.

Going forward: I will mount the encoders and work on getting a prototype over the weekend.

I worked on fusion designing an encoder case and a bolt that can attach to the motor.

https://github.falmouth.ac.uk/JH321562/Comp-208-Brab-Putt/commit/df50d0597a7f96f25dae85fd785dba7ac873e8c2

The measurements should be correct, but I'm preparing for the possibility that something may be slightly off. Everything in this design is friction fit and packed as tightly as possible to keep components from shifting. Without a PCB, every single placement has to be exact.

Because I wanted the wheels to cover most of the board, the available surface area is tiny, and fitting all the components has been a real challenge. Part of me feels like I should have done this redesign a week ago, but I wanted to be completely sure the final version would actually work before committing.

I knew going into this that redesigning so late was a risk... A step back with the hope of taking two steps forward. Right now, though, I'm still in the middle of that risk, and it's hard to tell whether it's paid off. It doesn't feel like it has yet, but maybe that's just part of the process.

DAY 42 30/11/2025
Goal: Finish wiring diagram
Outcome: couldn't finish wiring diagram so begun testing and assembling prototype.

Going forward: I will put together all the components tomorrow.
Tried to complete the wiring diagram on easy eda however due to a network error I couldn't sign in. However, I did get the prototype moving again and it’s looking quite exciting. I’m also preparing the full set up, utilizing a Arduino mega and 2 h-bridges this time.

DAY 43 2/12/2025
Goal: Drill exact holes in board
Outcome: Got board done, Motors fixed, and camera detecting.
Going forward: I will worked on the pixy cam in the background and begin implementing path finding.
Supervised lab with Ben, I got the chasis holes done exactly in the laser cutter, printed off more wheel connectors and demonstrated the prototype to Jareth while he worked on the object detection. Got the board fixed back together and completed the chasis.
DAY 44 3/12/2025
Goal: complete poster
Outcome: completed most of the wording for the poster.
Going forward: add more equations, add more to repo and overall clean up.
Worked on the poster all day documenting all features of the project and why we had made those choices, how the robot is meant to work and why it works. Why we chose those materials and how those choices impacted the project.

DAY 47 Day 6/12/2025
Goal: Rewire correctly prototype, finish chassis, and implement object detection.
Outcome: Rewire is complete, and code is now optimized. The chassis may be finished I don’t know how it will I will turn it on if it is under a box. Object detection was troublesome.
Going forward: I will continue to work on object detection; it works with the uno but not the mega.
Once I solve that problem, it will have an object following. Once I have the object, following it’s just setting pathfinding, then we’re done. Clean up rep, and that’s it
It is 15, nearly 16, and the chassis is done, now just to rewire it for Jareth’s code, object detection, object detection with movement and maybe something else
Got motors turning nicely and synchronised, so the pins are all correctly set now. Unfortunately, I couldn’t get the object detection on the Mega shown in the first video, but I could on the Uno. I don’t know why this is, but it is frustrating to deal with. I may have to ask Matt about it tomorrow.
DAY 48 07/12/2025
Goal: get motor moving from camera.
Outcome: motor moves without encoders
Going forward: chassis, repo, pathfinding, obstacle avoidance.
I took the encoder wiring out, and it worked. It moved and everything. I sent Jareths the video, and I’m very happy. Now just chassis, waterproofing, repo, pathfinding. On a 15-square track. Finalising code to go to red and avoid dark. Go to signature one and avoid signature two.
DAY 49 8/12/2025
Goal: Chassis is currently being printed. Jareth is working on the raspberry pi and getting it working for the field. Getting the repo ready with the video of the whole project. I also got the battery connected with strong cabling.
Outcome: Chassis is being built and I just don’t know how to integrate the original board with this new one.
Going forward: I will think about the repo all the other odd jobs to complete and just work with the all the components already made.
Finally got it tested working without a power supply. Unfortunately, I displayed it to matt without video presentation.
Because all I want is something to place on top. The chassis should be kept in place by the motors and tolerances made.
I’ve made a component to fit to the board so I think I’ve now solved that problem with the chassis.
I’ll have a look over at my devlog and video back home. Jareth’s now taking over and working hard with the project. I’ve provided him with the perfect.

DAY 50 9/12/2025

Goal: field test prototype and finalise design.

Outcome: 3D print worked well only thing holding it back was that I bolted the plate underneath instead of just keeping it all in the same print.

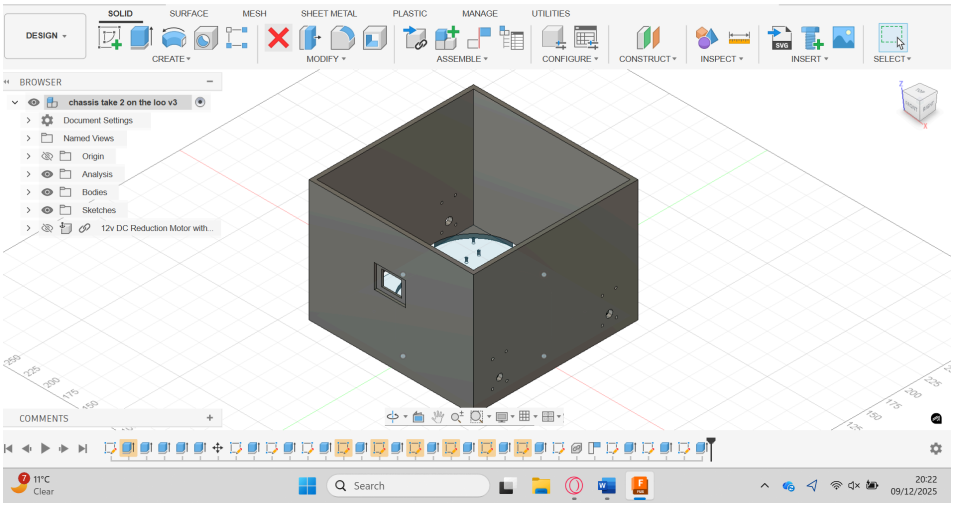
Going forward: I will try to plan further ahead making more sustainable designs. Make further modifications and work on code.

I managed to fix my slight design error by bolting the design in.

Jareth troubleshooted the device and after a field test at a new field, we found that the ground clearance is not enough. We need to put the wheels and motors lower. He designed new motor holders after I ran through some designs we could implement right away.

I worked on my poster and researched parts to help Jareths raspberry Pi issue:https://forums.raspberrypi.com/viewtopic.php?t=364311. He told he's already got the component so I trusted him and ensure that we would have a back up by modifying my backup version with the pixy cam.

I also managed to design an improved iteration on the box: https://github.falmouth.ac.uk/JH321562/Comp-208-Brab-Putt/blob/main/CAD/shell/chassis take 2.f3z



attachment:30ebe088-9f83-4f69-b987-58cbfeaf17ac:MicrosoftTeams-video_(4).mp4

AFTER ASSIGNMENT DEADLINE / after 16:00 on 12/12/2025

DAY 51 10/12/2025

Goal: Complete poster, with documentation and mathematics and format devlogs into something more organised.

Outcome: I got the poster at a point where I was satisfied with it however I did not implement what I had wanted to. I did not finish the video, or the readme.md.

Going forward: I will continue to clean up the repo.

DAY 52 11/12/2025

Goal: Help Jareth set up his new clearance motor holders and continue trouble shooting project. I managed to switch the pins from digital to PWM. Although the some motors weren't running and the battery needs a recharge.

Outcome: We stayed in the lab till 8:00 - I got there at 11:00. We uncounted a tripping issue with the robot that was resolved by the end by simply placing the motor holders diagonally.

Going forward: We will be able to implement this robot now into the field it just needs some modifications.

Such as path finding and a better searching algorithm.

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DAY 53 12/12/2025

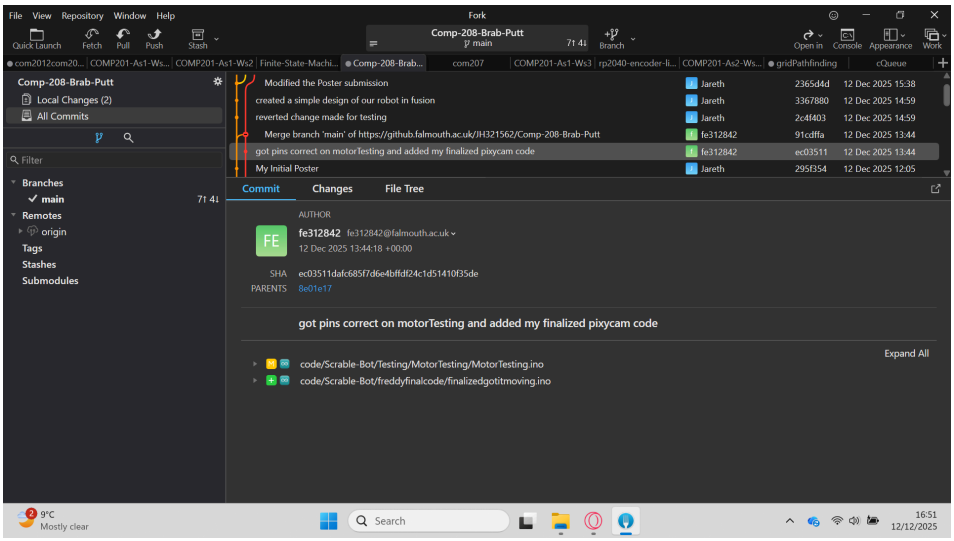
OFFICIAL DEADLINE for the module

Goal: manage to submit everything and try new wiring configuration. Now the encoder wiring module connects and it is more a fitted connection than the exposed wire and the pertruding male end of the motor.

Outcome: I only managed to do the encoders and finish my poster as well as add the research using the research I had from my poster.
Going forward: I think I will try this project again in my spare time using all of my parts and knowledge. I want to try and make a 2 wheeled taller version.

Unfortunately I committed code 2 hours before deadline but it didn't seem to be on the repo for what ever reason. It was important as well since it was the base of my poster talking about the state machine (which it had implemented) however without it, it questions the solidarity of my poster.

here is the evidence:



For whatever reason it isn't on the github repo but here is the code to demonstrate that I was honest in my poster.

```
#include <Pixy2.h>
#include <Pixy2CCC.h>

Pixy2 pixy;

// Motor pins
const int motor3a = 38;
const int motor3b = 40;
const int motor1a = 41;
const int motor1b = 39;
const int motor2a = 45;
const int motor2b = 43;
const int motor4a = 42;
const int motor4b = 44;
```

```
// Tracking variables
float deadzone = 0.15;
int signature, x, y, width, height;
float cx;
//state machine
enum State { SEARCH, TRACK_CAN, AVOID, CENTERED, TURN };
State state = SEARCH;

void setup()
{
  Serial.begin(115200);
  pixy.init();
  pixy.changeProg("color_connected_components");

  pinMode(motor1a, OUTPUT);
  pinMode(motor1b, OUTPUT);
  pinMode(motor2a, OUTPUT);
  pinMode(motor2b, OUTPUT);
  pinMode(motor3a, OUTPUT);
  pinMode(motor3b, OUTPUT);
  pinMode(motor4a, OUTPUT);
  pinMode(motor4b, OUTPUT);
}

// Movement functions
void forward() {
  digitalWrite(motor1a, HIGH); digitalWrite(motor1b, LOW);
  digitalWrite(motor2a, HIGH); digitalWrite(motor2b, LOW);
  digitalWrite(motor3a, HIGH); digitalWrite(motor3b, LOW);
  digitalWrite(motor4a, HIGH); digitalWrite(motor4b, LOW);
}

void backward() {
  digitalWrite(motor1a, LOW); digitalWrite(motor1b, HIGH);
  digitalWrite(motor2a, LOW); digitalWrite(motor2b, HIGH);
  digitalWrite(motor3a, LOW); digitalWrite(motor3b, HIGH);
  digitalWrite(motor4a, LOW); digitalWrite(motor4b, HIGH);
}

void stopMotors() {
  digitalWrite(motor1a, LOW); digitalWrite(motor1b, LOW);
  digitalWrite(motor2a, LOW); digitalWrite(motor2b, LOW);
  digitalWrite(motor3a, LOW); digitalWrite(motor3b, LOW);
  digitalWrite(motor4a, LOW); digitalWrite(motor4b, LOW);
}

// Mapping
float mapfloat(long x,long in_min,long in_max,long out_min,long out_max)
{
  return float(x - in_min) * (out_max - out_min) / float(in_max - in_min) + out_min;
}

// Read Pixy, return signature or 0 if none for debugging
```

```
int checkPixy()
{
  uint16_t blocks = pixy.ccc.getBlocks();

  if(blocks == 0) return 0;

  signature = pixy.ccc.blocks[0].m_signature;
  x = pixy.ccc.blocks[0].m_x;
  width = pixy.ccc.blocks[0].m_width;

  cx = x + (width / 2);
  cx = mapfloat(cx, 0, 316, -1, 1);

  return signature;
}

// AVOID BEHAVIOUR
void avoidObstacle()
{
  Serial.println("AVOIDING");

  // Stop then back up then turn right
  stopMotors();
  delay(150);

  backward();
  delay(250);

  // turn away from obstacle
  digitalWrite(motor1a, HIGH); digitalWrite(motor1b, LOW);
  digitalWrite(motor2a, LOW);  digitalWrite(motor2b, HIGH);
  digitalWrite(motor3a, HIGH); digitalWrite(motor3b, LOW);
  digitalWrite(motor4a, LOW);  digitalWrite(motor4b, HIGH);

  delay(350);

  stopMotors();
}

void loop()
{
  int sig = checkPixy();

  // No detections == SEARCH
  if(sig == 0)
  {
    state = SEARCH;
  }
  // Signature 2 = obstacle == AVOID
  else if(sig == 2)
  {
    state = AVOID;
  }
  // Signature 1 = can to chase
```

```
  else if(sig == 1)
  {
    state = TRACK_CAN;
  }

  // STATES AND THEIR ACTIONS
  switch(state)
  {
    case SEARCH:
      stopMotors();
      break;

    case AVOID:
      avoidObstacle();
      state = SEARCH;
      break;

    case TRACK_CAN:
      // if in the centre go forward
      if(cx > -deadzone && cx < deadzone)
      {
        forward();
      }
      // If off-centre, bias movement
      else if(cx < 0)
      {
        forward();   // left bias
      }
      else if(cx > 0)
      {
        backward();  // right bias
      }
      break;
  }

  delay(40);
}
```

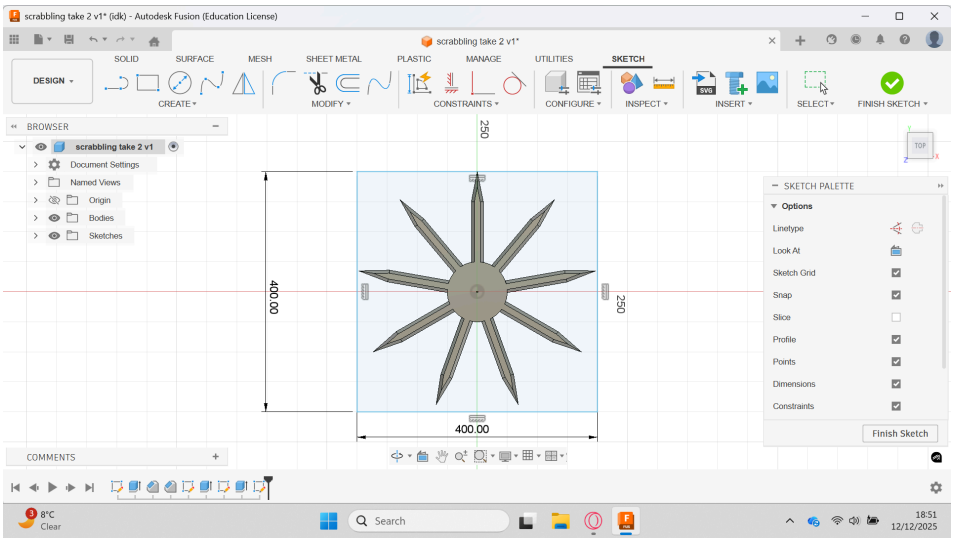
I also had the old poster on the github but my finalized oit's in the submission area. So hopefully that's enough.

Anyway it is now later in the day and I need to write this idea down.

So two big wheels attached to 2 small n20 motors, one arduino uno, one pixy cam cam and I think that is it. oh and one hbridge on and then 1 battery.

Use this for the wheels.

The most important thing would be balancing the arduino and keeping the camera flat inbetween the two wheels.



Then I went to begin simplifying Jareths worksheet 9 code:

